

ITC C103 FUNDAMENTALS OF IT WW2 - M2U4 Midterms Assignment 6

BSIT - 102I

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Answers:

1. You've been asked to create a digital contact list for your team. What are the key structural components of a database table you would use to organize information like names, phone numbers, and email addresses?

Answer: A contact table would use fields (columns) like name, phone number, and email address, with a primary key to uniquely identify each record and data type constraints to ensure proper formatting of information.

2. A manager asks you to generate a report showing all employees who work in the marketing department. What is the standard term for the process of pulling specific information from a database, and what language is commonly used to do this?

Answer: The process is called a query, and the standard language used is SQL (Structured Query Language), which allows you to retrieve specific data such as all marketing department employees.

3. Your colleague suggests using Microsoft Excel to store a large list of product inventory, while you propose using a Relational Database Management System (RDBMS) like Microsoft SQL Server. What are two practical reasons why the RDBMS is a better choice for this task?

Answer: RDBMS is better because it can efficiently handle millions of records without performance degradation like Excel would experience, and it enforces data integrity while allowing multiple users to access and modify data simultaneously without corruption.

4. A company's database has separate tables for Customers and Orders. How is the connection between a customer and their specific orders defined in the database, and why is this important for maintaining data integrity?

Answer: The connection is defined through a foreign key relationship where the Orders table contains a customer_id column that references the primary key in the Customers table, which maintains referential integrity by ensuring every order belongs to a valid customer.

5. You have a new dataset from a social media company. The data includes customer reviews, which are written in free-form text, and customer profiles, which are neatly organized in a spreadsheet. What are the technical terms for these two types of data, and how do they impact the way you would store and analyze them?

Answer: Customer reviews (free-form text): unstructured data and Customer profiles (organized in a spreadsheet): structured data. Structured data fits neatly into relational databases (rows/columns) and can be queried easily with SQL. Unstructured data doesn't fit a fixed format, it's often stored in NoSQL databases or data lakes, and requires special techniques (like text mining or natural language processing) to analyze.

6. You are a database administrator tasked with updating a list of employee salaries. Would you use a Data Definition Language (DDL) or a Data Manipulation Language (DML) statement to perform this action, and what is the difference between the two?

Answer: Updating existing salary values is a DML (Data Manipulation Language) task, specifically an UPDATE statement. DDL defines/changes the structure of the database (examples: CREATE, ALTER, DROP tables). DML manages the data inside the tables (examples: INSERT, UPDATE, DELETE, SELECT). Since you're changing salary values (data), not the table structure, you'd use DML.

7. Your company wants to build a new business application. Explain how a client-server architecture would function for this application, and how its design can be made more flexible for future updates and growth.

Answer: The client (e.g., a web browser or app on the user's device) sends requests. The server processes those requests (runs business logic, accesses the database) and sends back responses. This separation lets many clients connect to one central server for consistent data and logic.